

Week 5: Naive Bayes Classification

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Review classification task: Given a data set, $X \in \mathbb{R}^{N \times d}$ that has N rows and d dimensions, and $y_i \in \{C_1, C_2, \dots, C_k\}$ where C_k is a class k .

Use cases:

- Churn prediction
- Stock prediction
- Email classification (link)

Bayes' Rule

$$\underbrace{P(Y | X)}_{\text{Posterior}} = \frac{\underbrace{P(X | Y)}_{\text{Likelihood}} \cdot \underbrace{P(Y)}_{\text{Prior}}}{\underbrace{P(X)}_{\text{Marginal}}}$$

There are four parts:

- **Posterior probability** (the conditional probability of Y given X)
- **Prior probability** (the initial belief regarding the truth of a statement)
- **Likelihood** (the chance of observing a particular sample X when the parameter is equal to Y)
- **Marginal probability or Evidence** (the unconditional probability (overall populations))

Bayes' Rule (Conditional Probabilities)

$$P(Y|X) = \frac{P(Y \cap X)}{P(X)} \quad (1)$$

This says that conditional probability is the probability that both X and Y occur divided by the unconditional probability that X occurs.

$$P(X|Y) = \frac{P(X \cap Y)}{P(Y)} \quad (2)$$

from (2), we obtain

$$P(X|Y)P(Y) = P(X \cap Y) \quad (3)$$

We have known that

$$P(X|Y)P(Y) = P(X \cap Y) = P(Y \cap X) \quad (4)$$

from (1), we get

$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)} \quad (5)$$

Bayes' Rule Classification Example

X_1 (Name)	Y (Sex)
Drew	M
Claudia	F
Drew	F
Drew	F
Alberto	M
Karin	F
Nina	F
Sergio	M
Drew	?

Goal: Predict gender of the last "Drew" using Bayes' Rule.

$$P(Y_i = M \mid X_i = \text{Drew}) = \frac{P(\text{Drew} \mid M) \cdot P(M)}{P(\text{Drew})} = \frac{\frac{1}{3} \cdot \frac{3}{8}}{\frac{3}{8}} = \frac{1}{3}$$

$$P(Y_i = F \mid X_i = \text{Drew}) = \frac{P(\text{Drew} \mid F) \cdot P(F)}{P(\text{Drew})} = \frac{\frac{2}{5} \cdot \frac{5}{8}}{\frac{3}{8}} = \frac{2}{3}$$

$$\text{evidence} = \sum_{i=1}^K P(\mathbf{X} \mid y_i) \cdot P(y_i) \quad \text{where } K \text{ is the number of classes}$$

Multiple Features

X_1 (Name)	X_2 (Over 170cm)	X_3 (Eye Color)	X_4 (Hair Length)	Y (Sex)
Drew	No	Blue	Short	M
Claudia	Yes	Brown	Long	F
Drew	No	Blue	Long	F
Drew	No	Blue	Long	F
Alberto	Yes	Brown	Short	M
Karin	No	Blue	Long	F
Nina	Yes	Brown	Short	F
Sergio	Yes	Blue	Long	M

Example: Predict Y for X_1 =Drew, X_2 =No, X_3 =Brown, X_4 =Short

$$P(Y \mid x_1, x_2, x_3, x_4) = \frac{P(x_1, x_2, x_3, x_4 \mid Y) \cdot P(Y)}{P(x_1, x_2, x_3, x_4)}$$

Concept of Bayes' Rule

Assumption: Features (x_i) are dependent

For a joint distribution of d variables (with binary values), all $2^d - 1$ combinations must be known.

General form of likelihood (Chain rule of probability):

$$P(x_1, x_2, \dots, x_d \mid Y) = P(x_1 \mid Y)P(x_2 \mid x_1, Y)P(x_3 \mid x_2, x_1, Y) \dots \\ \dots P(x_d \mid x_{d-1}, x_{d-2}, \dots, x_2, x_1, Y)$$

Naive Bayes Assumption

Assumption: Attributes x_i are conditionally independent given class y .

Likelihood:

$$P(\mathbf{X} \mid y_i) = P(x_1 \mid y_i) \cdot P(x_2 \mid y_i) \cdot \dots \cdot P(x_d \mid y_i)$$

Example:

$$\begin{aligned} P(x_1, x_2, x_3, x_4 \mid Y) = \\ P(\text{name} = \text{Drew} \mid Y = M) \cdot P(\text{over 170} = \text{No} \mid Y = M) \\ \cdot P(\text{eye color} = \text{brown} \mid Y = M) \cdot P(\text{hair} = \text{short} \mid Y = M) \end{aligned}$$

Example computation

$$P(x_1 = \text{Drew}, x_2 = \text{No}, x_3 = \text{brown}, x_4 = \text{short} \mid \text{Male}) \cdot P(Y = \text{Male})$$

$$P(x_1 = \text{Drew} \mid \text{Male}) = 1/3$$

$$P(x_2 = \text{No} \mid \text{Male}) = 1/3$$

$$P(x_3 = \text{brown} \mid \text{Male}) = 1/3$$

$$P(x_4 = \text{short} \mid \text{Male}) = 2/3$$

$$= 2/81 \cdot P(Y = \text{Male}) = 2/81 \cdot 3/8 = 1/108 = 0.0092$$

Example computation

$$P(x_1 = \text{Drew}, x_2 = \text{No}, x_3 = \text{brown}, x_4 = \text{short} \mid \text{Female}) \cdot P(Y = \text{Female})$$

$$P(x_1 = \text{Drew} \mid \text{Female}) = 2/5$$

$$P(x_2 = \text{No} \mid \text{Female}) = 3/5$$

$$P(x_3 = \text{brown} \mid \text{Female}) = 2/5$$

$$P(x_4 = \text{short} \mid \text{Female}) = 1/5$$

$$= 12/625 \cdot P(Y = \text{Male}) = 12/625 \cdot 5/8 = 3/250 = 0.012$$

Final computation

Final answer is Male.

$$P(Y = \text{Male} \mid x_1, x_2, x_3, x_4) = \frac{0.0092}{0.0092 + 0.012} = \frac{0.0092}{0.0212} = 0.43$$

$$P(Y = \text{Female} \mid x_1, x_2, x_3, x_4) = \frac{0.012}{0.0092 + 0.012} = \frac{0.012}{0.0212} = 0.56$$

Advantages and Disadvantages

Advantages:

- Fast to train/classify
- Handles streaming data (e.g., Email spam detection)

Disadvantage:

- Assumes independence of features (X)
- Link: Is the naive independence assumption valid for all dataset?

Handling Continuous Values

Previous Example: $x_i \in \{\text{Categories}\}$ — Compute probability by counting.

If $x_i \in \{\text{Continuous-values}\}$, we will assume the values follow a Gaussian distribution.

Example: $x_i = \text{Height (cm)}$

Values: 100, 101, 99, 120, ..., 160

Gaussian Probability Density Function:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma_k^2}} e^{-\frac{(x-\mu_k)^2}{2\sigma_k^2}}$$

where μ_k and σ_k is belong to class k

where $e = 2.7183$

Laplacian Correction (Estimator)

Dealing with zero probability values:

Example: Training set contains 1000 samples

- Income = low: 0 samples, medium: 990 samples, high: 10 samples

Without Laplace:

$$P(\text{low}) = 0$$

With Laplace Correction:

$$P(\text{low}) = \frac{0 + 1}{1000 + 3}$$

Example

- Olabenjo, Babatunde. "Applying naive bayes classification to google play apps categorization." arXiv preprint arXiv:1608.08574 (2016).